

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING - CYBER SECURITY

REPORT OF TECHNICAL TALK ON “APPLICATIONS OF AI-ML IN CYBER SECURITY” Held on 29 April 2026

The technical talk on “Applications of AI-ML in Cyber Security”, held on 29th April 2026 at CS 401 Hall, was delivered by Dr. Pavan Kumar C, Assistant Professor from the Department of Computer Science and Engineering at Indian Institute of Information Technology, Dharwad. The event was organized for 4th and 6th semester students of CSE- Cyber Security.

The session began with an introduction to the rapid evolution of cyber threats in today’s digital era and the limitations of traditional security mechanisms in handling large-scale and sophisticated attacks. Dr. Pavan Kumar highlighted how the integration of Artificial Intelligence (AI) and Machine Learning (ML) has become essential in modern cyber security frameworks, enabling systems to automatically learn patterns, identify anomalies, and respond to threats in real time without constant human intervention. He explained key concepts such as supervised and unsupervised learning, emphasizing their roles in detecting intrusions, malware classification, phishing identification, and fraud detection.

The speaker provided detailed insights into practical implementations by showcasing various real-world datasets and platforms, including York Dataset and CIC Dataset, which are widely used in cyber security research and industry applications. He explained how such datasets help in training ML models to recognize malicious activities by analyzing network traffic, user behavior, and system vulnerabilities. Through examples, he illustrated how AI-based systems can proactively predict potential attacks and minimize risks by identifying unusual patterns before they escalate into major security breaches. The importance of data quality, feature selection, and model evaluation in building efficient cyber security solutions was also discussed in depth.

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In addition to technical concepts, Dr. Pavan Kumar stressed the importance of skill development and continuous learning in the cyber security domain. He guided students on various globally recognized certifications that are highly valued in the job market, explaining how certifications not only validate technical knowledge but also improve career prospects in roles such as security analyst, penetration tester, and cyber security engineer. He encouraged students to actively participate in hands-on projects, internships, and research activities to gain practical exposure and stay updated with the latest industry trends.

The session also included an interactive component where students were encouraged to ask questions and clarify their doubts regarding AI applications, career pathways, and project ideas in cyber security. Dr. Pavan Kumar provided thoughtful responses and shared valuable insights from his academic and professional experience. Overall, the talk was highly enriching and informative, bridging the gap between theoretical knowledge and real-world applications. It inspired students to explore the intersection of AI and cyber security, equipping them with a clear understanding of emerging technologies and motivating them to build a strong foundation for future careers in this rapidly growing field. Here are some of the snaps of the event.



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30/4/26.
Event co-ordinator.

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